```
maulik@maulik-VirtualBox:~$ clamscan -r /home
/home/maulik/.local/share/keyrings/login.keyring: OK
/home/maulik/.local/share/keyrings/user.keystore: OK
/home/maulik/.local/share/.converted-launchers: Empty file
/home/maulik/.local/share/zeitgeist/fts.index/postlist.baseA: OK
/home/maulik/.local/share/zeitgeist/fts.index/position.baseB: OK
/home/maulik/.local/share/zeitgeist/fts.index/termlist.baseB: OK
/home/maulik/.local/share/zeitgeist/fts.index/termlist.baseA: OK
/home/maulik/.local/share/zeitgeist/fts.index/position.DB: OK
/home/maulik/.local/share/zeitgeist/fts.index/termlist.DB: OK
/home/maulik/.local/share/zeitgeist/fts.index/record.baseA: OK
/home/maulik/.local/share/zeitgeist/fts.index/record.baseB: OK
/home/maulik/.local/share/zeitgeist/fts.index/postlist.DB: OK
/home/maulik/.local/share/zeitgeist/fts.index/postlist.baseB: OK
/home/maulik/.local/share/zeitgeist/fts.index/iamchert: OK
/home/maulik/.local/share/zeitgeist/fts.index/record.DB: OK
/home/maulik/.local/share/zeitgeist/fts.index/position.baseA: OK
/home/maulik/.local/share/zeitgeist/fts.index/flintlock: Empty file
```

Figure 6: Scanning a particular directory

```
----------- SCAN SUMMARY -----------
Known viruses: 5723186
Engine version: 0.99.2
Scanned directories: 80
Scanned files: 90
Infected files: 0
Data scanned: 1.26 MB
Data read: 0.92 MB (ratio 1.37:1)
Time: 21.941 sec (0 m 21 s)
maulik@maulik-VirtualBox:~$
```

Figure 7: Scanning results

```
maulik@maulik-VirtualBox:~
maulik@maulik-VirtualBox:~$ crontab -e
no crontab for maulik - using an empty one

Select an editor.  To change later, run 'select-editor'.
  1. /bin/ed
  2. /bin/nano        <---- easiest
  3. /usr/bin/vim.tiny

Choose 1-3 [2]:
```

Figure 8: Scheduling a *cron* job

```
maulik@maulik-VirtualBox:~
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirecte
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cro
#
# m h  dom mon dow   command
0 1 * * clamscan -r /home
-- INSERT --
```

Figure 9: Modified *crontab* file

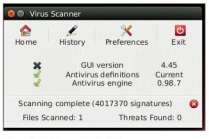

Figure 10: GUI prompt ClamTK

Figure 3 indicates an error while updating the virus database. This is because after the installation of the ClamAV daemon, the *freeclam* process is already running; so we need to stop it or kill it before running the command again.

The first time, it will take longer to update the database because it is freshly installed.

Now we will scan the */home* directory using ClamAV. Run the command given below to perform the scanning:

```
clamscan -r /home
```

By default, it will update the *freshclam* daemon every hour (24 times). We can change this by using the command given below:

```
sudo dpkg-reconfigure clamav-freshclam
```

To check the version of ClamAV, use the following command:

```
clamdscan –V
```

We can also set a *cron* job for it, so that it will repeatedly scan the mentioned drive/directory as per the given time.

To do that, run the command given below:

```
crontab -e
```

A *crontab* file will be opened and you can append the file in the given (below) link before saving the file:

```
0 0 1 * * clamscan -r /location
```

It will run the ClamAV every first day of the month at midnight (12 a.m.).

Here is the *crontab* format for reference:

```
Minute   Hour   Day of Month   Month   Day of Week        Command
(0-59)   (0-23)   (1-31)   (1-12 or Jan-Dec)  (0-6 or Sun-Sat)
   0        0         1
   *        *         *
clamscan -r  /home
```

Till now, we have seen scanning using CLI; we can do the same thing using a GUI too. Run the command given below to do so:

```
sudo apt-get install ClamTK
```

If you find any difficulties during this step, please refer to the link given below for troubleshooting.

http://askubuntu.com/questions/378558/unable-to-locate-package-while -trying-to-install-packages-with-apt

Similarly, we can install ClamAV for Windows. Refer to the link that follows to download the *.msi* file for the Windows version.

https://www.clamav.net/downloads

There are many third party tools supported by ClamAV, though ClamAV itself does not provide any support for those tools.

You can refer to the official site of ClamAV at *https://www.clamav.net* for more information.

By: Maulik Parekh

The author has an M. Tech degree in cloud computing from VIT University, Chennai. He can be reached at *maulikparekh2@gmail.com*. Website: *https://www.linkedin.com/in/maulikparekh2*